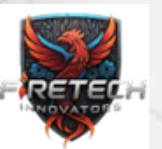


BUILDING THE FUTURE OF CITIES

Sustainable Infrastructure for
Climate-Resilient Urban Transformation



Produced by Dhurgham Alsaadi, 2026



The Problem: Cities at Risk

1.6 BILLION

people lack adequate housing

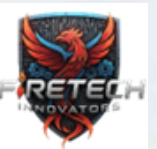
Rising to **3 BILLION** by 2030

Cities like Lagos, Jakarta, and Dhaka face extreme climate impacts

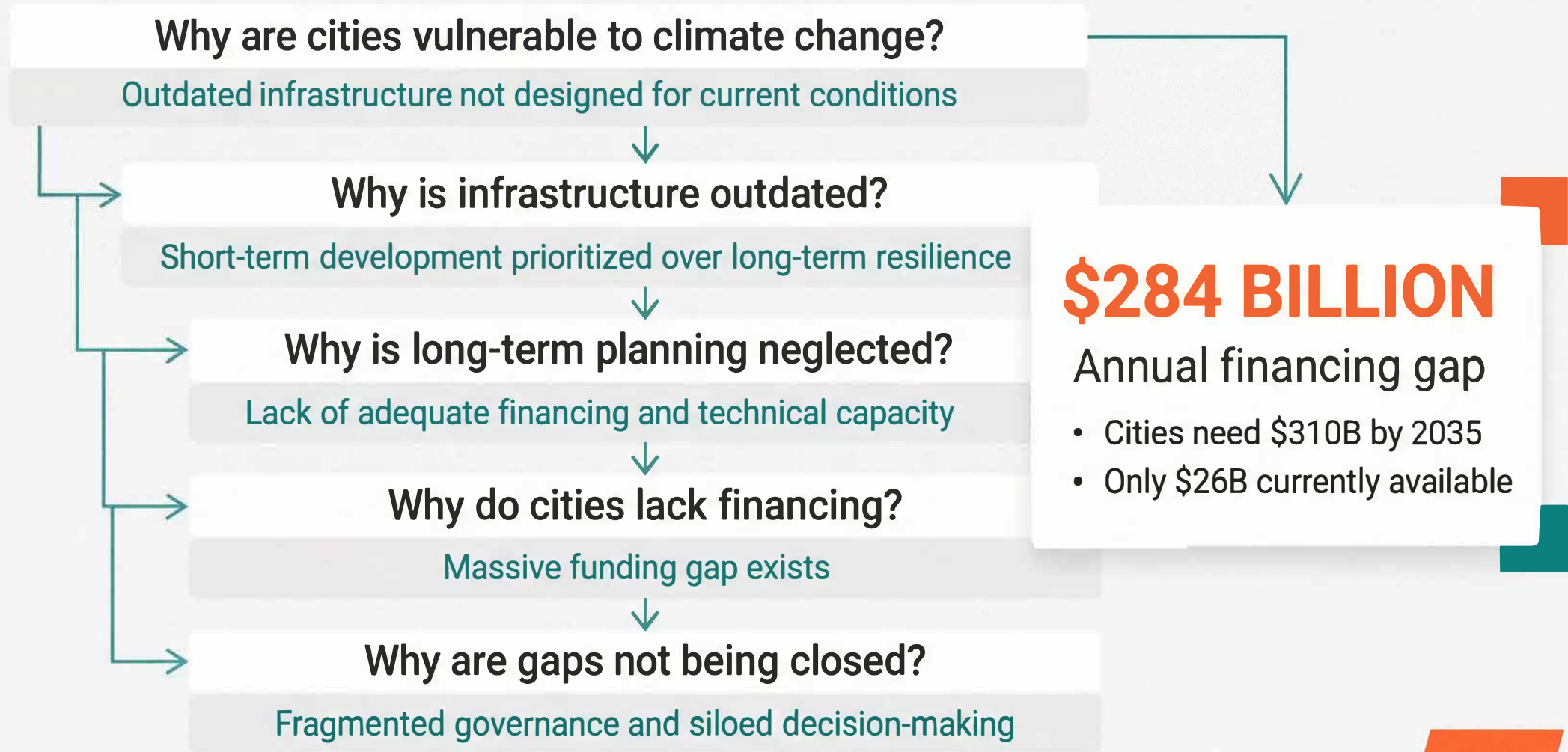
Residents in informal settlements experience:

- Rising sea levels and intense storms
- Urban heat islands
- Inadequate infrastructure
- Limited resources for adaptation

Nearly every urban resident is exposed to climate change, but vulnerable communities are disproportionately affected



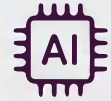
Root Cause: The 5 Whys Analysis



Core Problem: Systemic failure in urban governance, finance, and infrastructure design

Solution: Integrated Smart Infrastructure

A unified digital and physical ecosystem using IoT and AI to optimize energy, water, and waste flows in real time

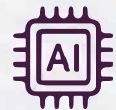


Energy Optimization

- Smart grids with IoT sensors
- AI algorithms for dynamic adjustment

74% reduction in transmission losses

40% reduction in grid congestion



Water & Waste Management

- Real-time monitoring systems
- Predictive leak detection

30% reduction in non-revenue water

25% reduction in waste collection costs

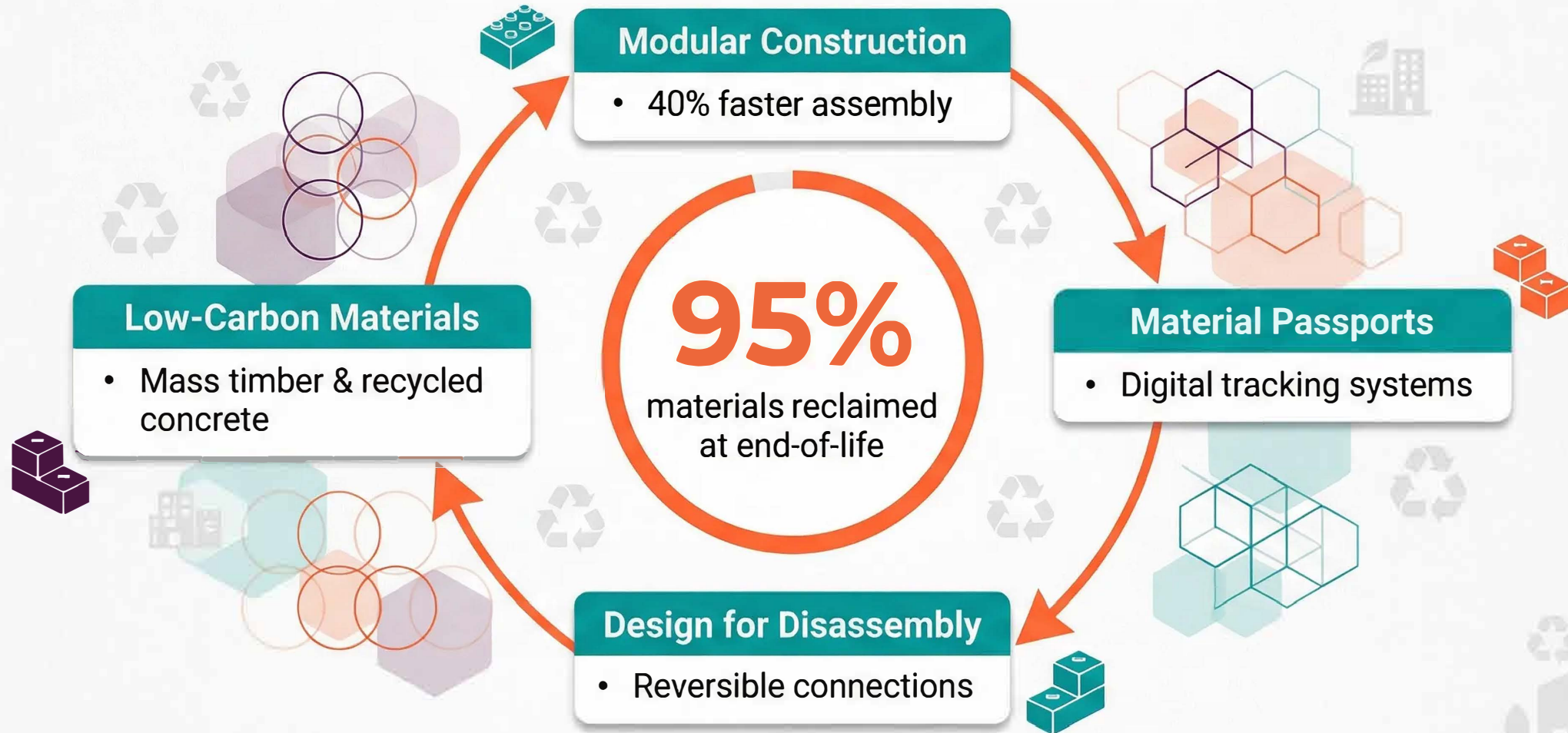


Climate Resilience

- Predictive weather models
- Proactive intervention systems
- Pre-flood pumping strategies
- Heatwave cooling optimization



Solution: Circular Design Principles



Waste-to-Resource Loop: Construction waste becomes new building materials - crushed concrete for green roofs, reclaimed wood for community gardens

Market Opportunity & Urban Trends

Market Segment	2025 Value	2030/2034 Forecast	CAGR
Smart Cities Market	\$850.565B	\$1.91T	17.65%
Green Buildings Market	\$565B	\$1.374T	9.42%
Sustainable Infrastructure Market	\$71.04B	\$405.08B	12.4%

70%

of global population will live
in cities by 2050



1.3 BILLION

urban population growth
expected in Africa alone



Target Markets: Rapidly urbanizing cities in emerging economies

- Africa, South Asia, and Latin America
- Cities with net-zero commitments
- Active public-private partnerships
- Access to international climate finance



Revenue & Business Model

B2G Infrastructure Contracts



- Multi-year agreements (3-5 years)
- Design & management of smart infrastructure
- Performance-based incentives

\$50M projected revenue by Year 3

SaaS City Dashboard



- Real-time monitoring & analytics
- Tiered pricing: \$15K-\$150K/year
- Predictive compliance tracking

30% gross margin

Green Certification Services



- End-to-end sustainability support
- \$50K-\$200K per certification cycle
- Third-party verification

40% gross margin

Diversified revenue streams ensure financial stability and scalable growth

Traction & Validation

Pilot Success

Lagos & Jakarta Pilots 

- **30%** energy reduction
- **25%** waste cost savings
- **22%** peak load reduction

Xiong'an New Area

350,000 sqm deployment

40%

grid congestion reduction

- ✓ 38% flood risk reduction
- ✓ 94% satisfaction rate

Verified by China Academy of Urban Planning

Global Partnerships

MOUs with:

- World Bank
- International Finance Corporation
- Delphos

\$1.2B pipeline 
across 12 cities

Competitive Differentiation



Full Lifecycle Design

Cradle-to-cradle approach
unlike competitors who focus
on isolated components

Unified platform integrating
smart infrastructure, circular
construction, and AI planning



Material Passports

Digital tracking of every
building component's origin,
composition, and end-of-life
pathways

Enabling 95% material recovery
and reuse



AI-Powered Predictive Maintenance

Machine learning predicts
infrastructure failures before
they occur

40%

reduction in downtime

Positioned as premium, high-impact provider enabling measurable environmental, economic, and social returns

Financial Projections & Unit Economics

Unit Economics

Cost per Module: \$250,000

Revenue per Module: \$750,000

Gross Profit: \$500,000

Gross Margin: 67%

3-Year Revenue Projection

Year 1: 10 modules → **\$7.5M**

Year 2: 25 modules → **\$18.75M**

Year 3: 50 modules → **\$37.5M**

Key Financial Metrics

Break-Even: Month 28

EBITDA (Year 3): \$12M

Revenue CAGR: 125%

Scalable urban module deployments driving sustainable growth



Leadership Team & The Ask

Leadership Team

Dr. Farhad Alsaadi

Chief Executive Officer

Co-Founder

UNIG, FireTech Innovators

UNIG, FireTech Innovators

UNIG, FireTech Innovators

Leadership Team

Dhurgham Alsaadi, CEO

- First Year Computer Science Student at UNIG.
- Founder of FireTech Innovators.

The Investment Ask

\$1 MILLION INVESTMENT

Fund Allocation:

- 40% R&D Enhancement
- 30% Pilot Project Scaling
- 20% Team Expansion
- 10% Marketing & Partnerships

Projected Impact by Year 3:



100,000 Urban
Residents
Served



500,000
Tons CO₂
Reduced



\$50M Green
Investment
Mobilized

Contact Information & References

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Scan Me!

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Thank you for your time and consideration

